Case Study 1:

Non-invasive estimation and deadbeat control of pulsatile flow in an implantable rotary blood pump for heart failure patients.

References

- The research findings presented in these slides were published as follows:

[1] N. Malagutti, D. M. Karantonis, S. L. Cloherty, P. J. Ayre, D. G. Mason, R. F. Salamonsen, and N. H. Lovell," Noninvasive average flow estimation for an implantable rotary blood pump: a new algorithm incorporating the role of blood viscosity," *Artificial Organs,* vol. 31, pp. 45-52, 2007.

[2] A. H. AlOmari, A. V. Savkin, D. M. Karantonis, E. Lim, and N. H. Lovell, "Non-invasive estimation of pulsatile flow and differential pressure in an implantable rotary blood pump for heart failure patients," *Physiological Measurement*, vol. 30, pp. 371-386, 2009.

[3] A. H. AlOmari, A. V. Savkin, D. M. Karantonis, E. Lim, and N. H. Lovell," A dynamical model for pulsatile flow estimation in a left ventricular assist device," *BIOSIGNALS: Proc. Of 2nd Int. Conf. on Bio-inspired Systems and Signal Processing (Porto, Portugal), vol. 2, pp. 402-405*, 2009.

[4] E. Lim, A. H. AlOmari, A. V. Savkin, and N. H. Lovell," Noninvasive deadbeat control of an implantable rotary blood pump: A simulation study," *in Proc. Of 31st Annual. Int. Conf. of the IEEE Engineering in Medicine and Biology Society, Minneapolis, USA,* 2009, pp. 2855 - 8.

[5] E. Lim, S. Dokos, S. L. Cloherty, R. F. Salamonsen, D. G. Mason, J. A. Reizes, and N. H. Lovell."Parameter-optimized model of cardiovascular-rotary blood pump interactions," *IEEE Transactions on Biomedical Engineering*, Vol. 57, No. 2, 2010.

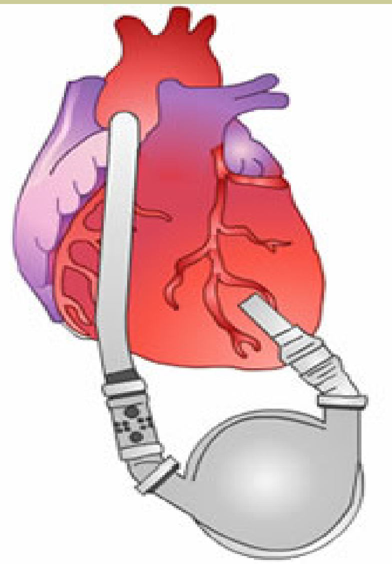
Research Projects

- Sensorless estimation of pulsatile flow and differential pressure (head) in an implantable rotary blood pump (IRBP) using non-invasive measurements of power and rotational pump speed as inputs.
- Developing a deadbeat control system of the pump flow coping with changing physiological demands.
- Validate and test the obtained dynamical models and controller using lumped parameter cardiovascular model, mock loop, animal experiments and then heart failure patients.

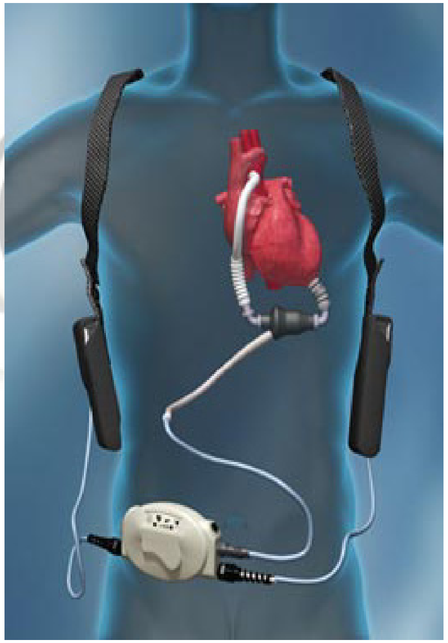
Background

- Congestive heart failure is a health condition where the heart is unable to work properly and unable to pump enough blood to cope with the body's physiological demands.
- The lack of donor organs for heart transplantation has led to a range of treatment options for heart failure patients.
- Implantable rotary blood pumps (iRBPs) are emerging as a viable long-term treatment alternative for heart failure patients. They used as bridge-to-transplant and hold the potential to become a long-term option to donor heart transplantation and destination therapy devices.
- A robust control system for pulsatile continuous flow of left ventricular assist devices (LVADs) which automatically responds to physiological cardiac demands and perturbations does not exist.

Motivations



- Currently, most available 2nd and 3rd generation of RBPs operate at a fixed target speed chosen by clinicians.
- Non-invasive estimation of pulsatile flow in an iRBPs is much less frequently studied.
- Sensorless instantaneous estimation of pulsatile flow and differential pressure (head) for iRBPs is crucial in designing an automatic, robust and responsive control system that effectively controls the flow according to the body's physiological needs and perturbations.
- It has been shown that flow rate and head can be accurately estimated under steady-state and non-pulsatile conditions.



Objectives

- To develop new, simple and elegant dynamical models that non-invasively estimate the pulsatile flow and head of an implantable rotary blood pump using feedback measurements of rotational pump speed and motor power.
- When operated in steady state conditions, the proposed dynamical pulsatile flow model can provide an accurate estimate of the flow which agrees with those obtained from a static model developed and verified in our laboratory under non-pulsatile conditions.
- Another practically important requirements of the presented dynamical models are their stability.
- To accurately study and estimate the transient response and the dynamics of the pulsatile flow and head.
- The dynamical models were used to develop a deadbeat controller for pulsatile pump flow in an IRBP.

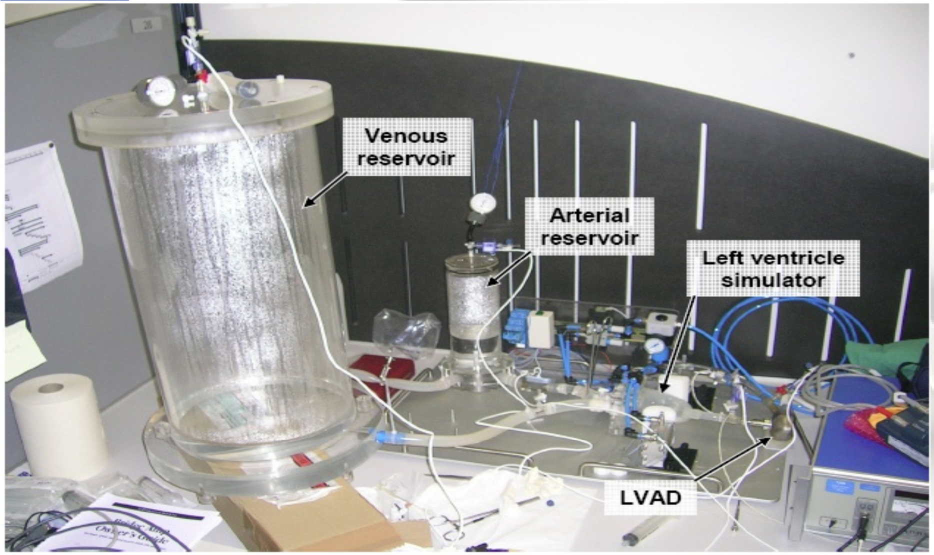
Materials and Methods

Non-invasive estimation of pulsatile flow and differential pressure in an IRBP

1) Pulsatile flow experiments:

- *In vitro* circulatory mock loop experiments:

- A VentrAssist®™ (Ventracor Limited, Sydney, Australia) left ventricular assist device (LVAD) was connected in a circulatory mock loop in a pulsatile environment as shown in figure (1) [2].



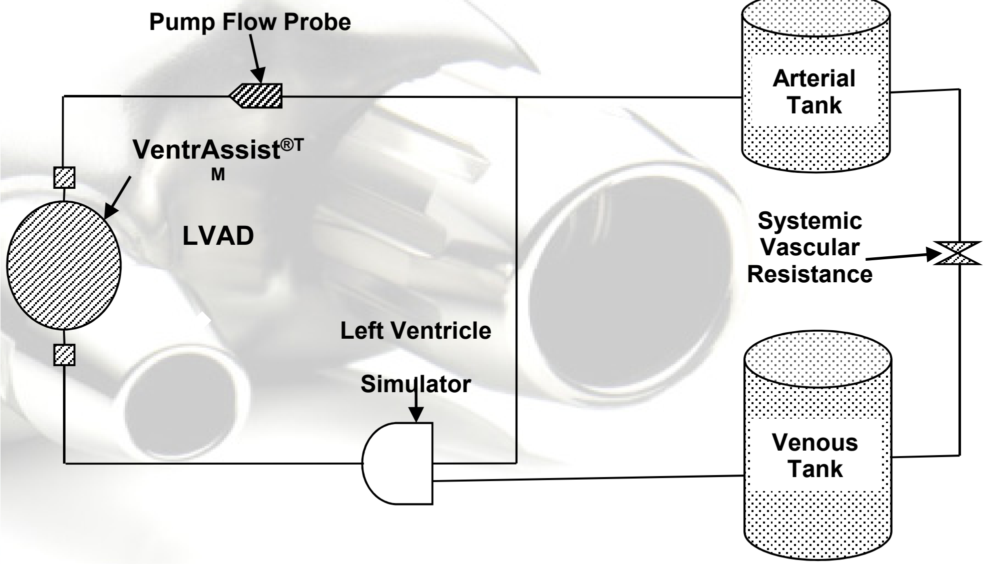


Figure (1):The pulsatile mock loop (left) and schematic diagram (right) used to generate the experimental data for flow and head.

Materials and Methods (Cont'd)

-The loop was composed of venous and arterial reservoir tanks, a silicone bag which represented the left ventricle chamber.

-Ventricular contraction was simulated by periodically compressing the mock ventricle with pneumatic pistons at a fixed rate.

-Arterial pressure, central venous pressure, pump inlet and outlet pressure were measured.

- The experiments were performed using six aqueous solutions of differing viscosity to cover the range of real human blood viscosities.

- This approximately coincides with hematocrit (HCT) values of 24, 27, 34, 39, 41, and 47.5 %.

Pulsatile flow experiments (Cont'd)

- To cover the entire operation range of the pump flow from -2 to 12 L/min with different viscosities, rotational speed was varied from 1800 to 3000 rpm in a stepwise increment of 100 rpm lasting for 30 s.

- In each step, the systemic resistance was also changed to cover the range of desired flows.

- The instantaneous rotational pump speed, motor voltage and current were accessed as a feedback signals from the pump controller.

-The sampling rate was 4 kHz for data recordings, but, in further analysis, the data were down-sampled to 50 Hz.

Pulsatile flow experiments (Cont’d)

- *Ex vivo* pigs experiments:

-The VentrAssist®™ LVAD was acutely implanted in six healthy pigs.

- In each pig, the inflow pump cannula was inserted into the apex of the left ventricle while the outflow cannula was anastomosed to the ascending aorta.

- Special transducers were instrumented to the pig’s native heart to measure left ventricle, left arterial, aortic, and pump inlet pressure.

- Pump and aortic flows were also measured from the outlet cannula of the pump and aorta respectively.

- The non-invasive rotational speed, motor current, and supply voltage were acquired from the pump controller.

- The experimental data were sampled at 200 kHz.

Materials and Methods (Cont'd)

2) Dynamical Modelling:

Pulsatile flow:

❖ Static flow model:

- A non-invasive, steady state average flow (Q_{ss}) estimator was designed in a non-pulsatile environment.
- The estimator was based on the motor's pump input power (VI) and rotational speed (ω) as follows [1]:

$$Q_{ss} = a + bVI + cVI^2 + dVI^3 + e\omega + h\omega^2, \quad (1)$$

-Where a, e, and h are constants and the power coefficients b, c, and d were described as having a linear relationship with HCT values.

❖ Dynamical pulsatile flow model:

- Here, we describe a dynamical flow model for the iRBP.
-The main requirement for the dynamical model is that any steady-state solution of the dynamical model be a solution of the static model (1).
- Furthermore, any steady-state solution of the dynamical model should be stable.
- We introduce a variable $f(k)$ as follows [2]:

Materials and Methods (Cont'd)

$$f(k) = g(P(k), \omega(k)), \qquad (2)$$

Where

$$g(P(k), \omega(k)) = a + bP(k) + cP^2(k) + dP^3(k) + e\omega(k) + h\omega^2(k). \qquad (3)$$

- Here $P = VI$ and k is the discrete time satisfying $t = kh$, $h > 0$ is the sampling interval.

- In other words, the variable $f(k)$ represents the right-side of the static equation (1), describing the pump flow in a steady state.
- We introduce a dynamical model of the form [2]:

$$(A(\nabla) + B(\nabla))Q_{est}(k) = B(\nabla)f(k),$$

Where $Q_{est}(t)$ is the output of the system represents the estimated instantaneous values of the pulsatile flow, $f(t)$ is the input to the dynamical system model, ∇ is the shift operator, $A(\nabla)$, $B(\nabla)$ are polynomials defined as follows:

Materials and Methods (Cont'd)

$$A(\nabla)Q_{est}(k)=\sum_{i=0}^{n}a_iQ_{est}(k-i), \quad (5)$$

$$B(\nabla)f(k)=\sum_{j=1}^{m}b_jf(k-j) \quad (6)$$

-Here n is the model output order, and m is the model input order satisfying $m \leq n$.

-Now, we describe all steady-state or constant solutions of equation (3).

-Let $Q_{est}(t) \equiv Q_0$ and $f(t) \equiv f_0$ for all k. Then, we obtain from (4) that [2,3]:

$$(A(1)+B(1))Q_0 = B(1)f_0. \quad (7)$$

- We assume $A(1)=0$ and $B(1)\neq 0$. This yields to the following conditions on parameters coefficients of the model:

$$\sum_{i=0}^{n} a_i = 0, \quad (8)$$

$$\sum_{j=1}^{m} b_j \neq 0. \quad (9)$$

Materials and Methods (Cont'd)

- Under the assumptions (8) and (9), it immediately follows from (7) that

$$Q_0 = f_0. \qquad (10)$$

- Since $f(k)$ is defined by (2),(3), and (10) it implies that Q_0 is the corresponding solution of the equation (1) [2].
- Hence, steady-states of the dynamical model are described by the static model.
- In other words, under the assumptions (8) and (9), the set of steady states of the dynamical model (4) coincides with the set of solutions of equation (1). Hence, the steady states of the dynamical model are described by the static model [2,3].
- Furthermore, if the system is stable i.e.; all $z : A(z)+B(z)=0$ belongs to $|z|<1$ (all poles of the system are inside the unit disk), then the solution of the dynamical system (4) with any initial conditions and with constant input f_0 will converge to the constant output Q_0 satisfying (10) and for any bounded input $f(k)$, a bounded flow output $Q_{est}(k)$ will result.

Materials and Methods (Cont’d)

Differential pressure (Head):

- We used an ARX model to derive a stable model for head estimation using as inputs the estimated pulsatile flow values and rotational speed.
- A discrete time ARX model with two input signals can be described as follows:

$$C(\nabla)H_{est}(k) = D_1(\nabla)Q_{est}(k) + D_2\omega(k), \quad (11)$$

Where

$$C(\nabla) = 1 + c_1\nabla^{-1} + c_2\nabla^{-2} + \cdots + c_n\nabla^{-n}, \quad (12)$$

$$D_1(\nabla) = d_{11}\nabla^{-1} + d_{12}\nabla^{-2} + \cdots + d_{1m}\nabla^{-m}, \quad (13)$$

and

$$D_2 = d_{21}\nabla^{-1} + d_{22}\nabla^{-2} + \cdots + d_{2m}\nabla^{-m}. \quad (14)$$

- Here, $H_{est}(k)$ is the estimated head, $Q_{est}(k)$ is the estimated pulsatile flow, $\omega(k)$ is the impeller rotational speed of the pump, n is the model output order and m is the input model order.

Materials and Methods (Cont'd)

3) System Identification and data analysis:

- The mock loop data were divided into 2 sets: one set was used for system identification while the other was used along with the animal data *(N = 6)* to validate the proposed models.
- The first set of data consisted of three experiments corresponding to the solution viscosities of 2.05, 2.66 and 3.26 mPas, while the other set contained data from the other three experiments with viscosities of 2.35, 2.95 and 3.56 mPas together with all the animal experimental data.
- In both, identification and validation data sets, the changes in the pump target speed were included in the data so that the transient response of the pump flow and head could be identified.
- In the system identification process, an off-line restricted least

Materials and Methods (Cont'd)

Least-squares method was used to estimate the parameter coefficients of pulsatile flow model while a least squares method was used to estimate coefficients for the head estimation model. Values of parameter coefficients of the pulsatile flow estimation model were chosen so that the error between estimated $Q_{est}(k)$ and measured flow $Q_{meas}(k)$ was minimised. Simultaneously, however, the resulting flow model parameter coefficients should fulfil the assumptions defined previously in equations (7) and (8).

- Similarly, in the head modelling, we divided the head data into two sets; one was used for system identification while the other set was used to generate $Q_{est}(k)$ using the derived flow estimation model and this was used with the corresponding values of the rotational speed $\omega(k)$ as inputs to the head model to estimate instantaneous values of $H_{est}(k)$. Also, parameter coefficients of the head estimation model were selected to minimise the error between the estimated measured head ($H_{meas}(k)$).

Materials and Methods (Cont'd)

-In both cases the model output order *(n)* and the input order *(m)* were varied from 1 to 10 and the delay value was determined by estimating the impulse response of the system by cross-correlation analysis between the input (s) and output signals, with a value of 1 used for both flow and head estimation models.

-The delay value was also set to a value of 1 when the pulsatile flow estimation model was used to estimate the flow in the animal experiments. The mean absolute error *(e)*, and correlation coefficient *(R)* between Q_{est}, Q_{meas} and H_{est}, H_{meas} were used to check the performance, and to evaluate the estimation accuracy of the models.

-The results obtained when using either of the two sets of data for identification and the other for validation attained the same degree of accuracy for both the pulsatile flow and head estimation tasks.

Dynamical Modelling Results

Pulsatile flow:

❖ Mock loop:

- Best results were obtained with system model orders of $n = 3$, $m = 2$, and $delay\ value = 1$. This combination gave the least mean absolute error (e) and highest correlation coefficient (R^2) between the estimated and measured flow and satisfied the conditions in equations (7) and (8).
- The resulting system model is described by the following difference equation [2]:

$$a_0 Q_{est}(k) + (a_1 + b_1) Q_{est}(k-1) + (a_2 + b_2) Q_{est}(k-2) + a_3 Q_{est}(k-3) = b_1 f(k-1) + b_2 f(k-2), \quad (15)$$

-Values of the estimated parameter coefficients are $a_0 = 1$, $a_1 = -2.2562$, $a_2 = 1.4950$, $a_3 = -0.2397$, $b_1 = 0.2710$, $b_2 = -0.2546$. $Q_{est}(k)$ is the estimated pulsatile flow and $f(k)$ is the input signal.

- The poles-zeros plot of the system model shown in figure 2 demonstrates that all poles are inside the unit circle, satisfying the stability of the model criteria.

Dynamical Modelling Results (Cont'd)

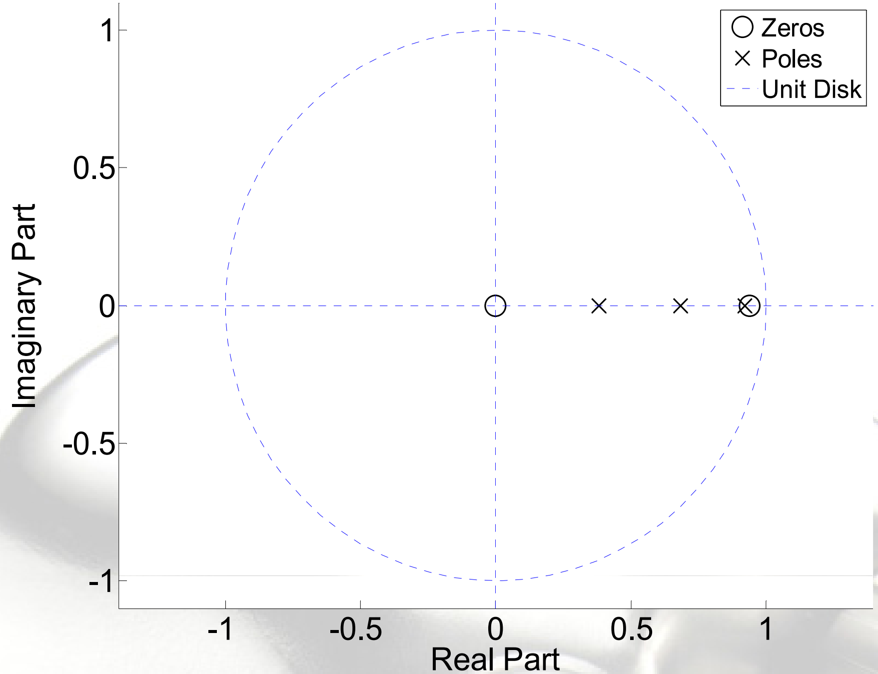


Figure (2): Poles-zeros plot of the system model described in equation (15). The system is stable as all poles are inside the unit circle.

-The dominant pole and a zero are close in value but are not exactly the same; there is a 5% difference in the absolute value between them. The reduced order model gave a high and clinically unacceptable mean absolute error *(e)* with low correlation coefficient (R^2) between estimated and measured pulsatile flow in both mock loop and animal experiments.

Dynamical Modelling Results (Cont'd)

-The dashed line in figure 3 (left and right panel) shows part of the estimated pulsatile and recorded measured flow together with the transient response of the pulsatile flow estimation model obtained from one of the mock loop experiments.

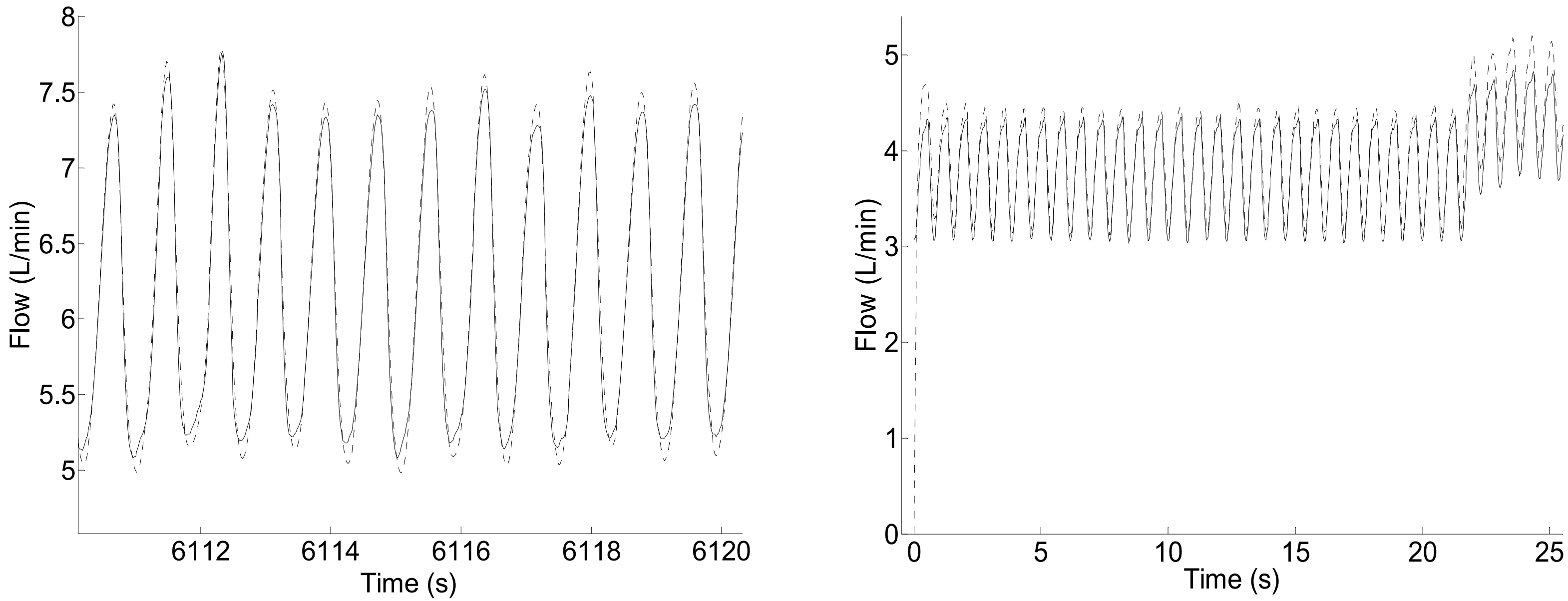


Figure (3): Estimated pulsatile flow compared with measured flow obtained from mock loop experiments at *w* = 2800 rpm and HCT = 27% (equivalent to viscosity value of 2.35 mPas) (left panel). Transient response of the pulsatile flow estimation model described in equation (17) from initial conditions to steady state and also the model response for target speed changes from $\omega = 1800\ rpm$ to $2000\ rpm$ with $HCT = 37\%$ (equivalent to viscosity value of $2.95\ mPas$) at time $21.6\ s$ (right panel). Solid line is measured flow (Q_{meas}), while dashed line shows estimated flow (Q_{est}).

Dynamical Modelling Results (Cont'd)

-Linear regression analysis between $Q_{est}(k)$ and $Q_{meas}(k)$ obtained from the circulatory pulsatile mock loop experiments are show in figure (4a-c).

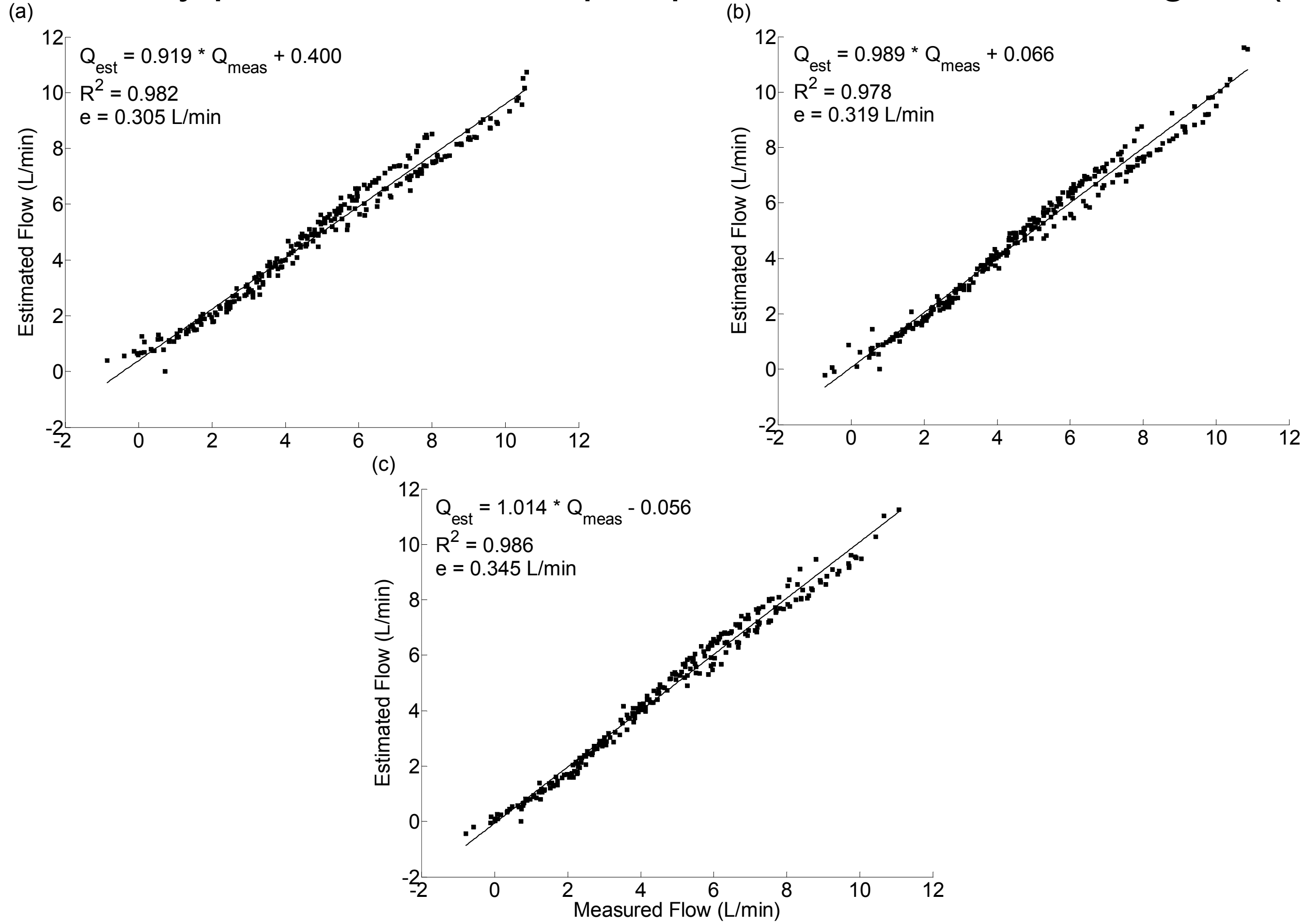


Figure (4): (a) $HCT = 27\%$ (equivalent to $2.35\ mPas$ viscosity), (b) $HCT = 37\%$ (equivalent to $2.95\ mPas$ viscosity) and (c) $HCT = 47.5\%$ (equivalent to $3.56\ mPas$ viscosity).

Dynamical Modelling Results (Cont'd)

❖ Animal experiments:

- During animal experimentation the values of HCT were not measured. Instead, for each animal we used the values of HCT that gave the best fit between the estimated and measured flows in the steady state model described in equation (1).
- The comparison between estimated pulsatile and measured flow together with the transient response of the pulsatile flow estimation model described in equation (15) in one animal experiment is shown in figure 5.

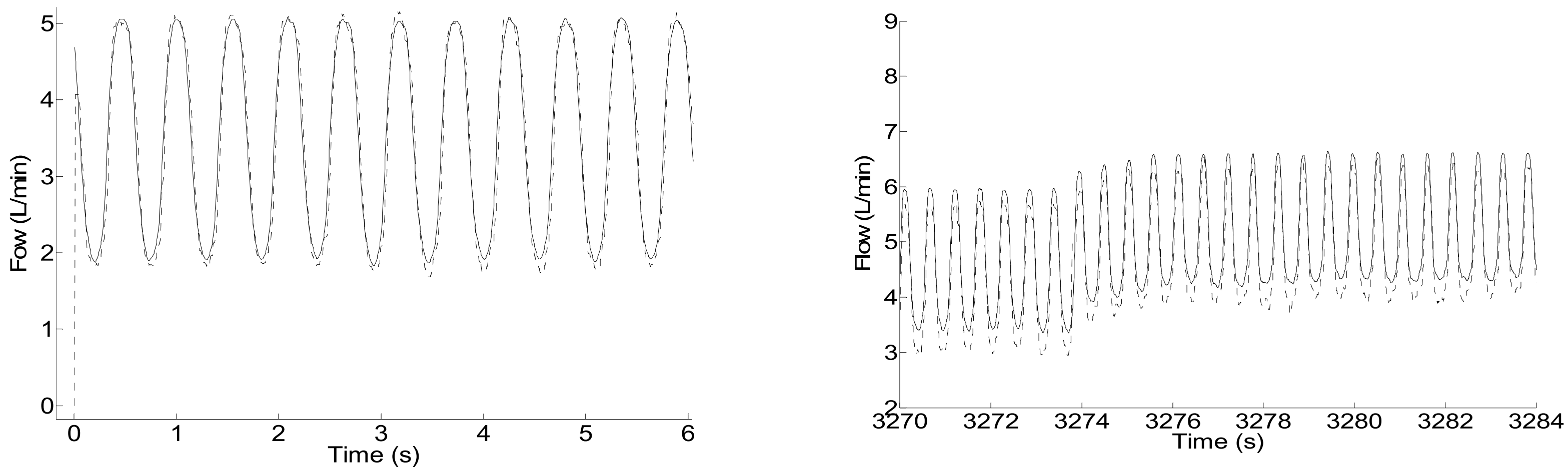


Figure (5): Estimated pulsatile flow (Q_{est}) compared with measured flow (Q_{meas}) in one animal experiment at $\omega = 1800$ rpm (left panel). Transient response of pulsatile flow estimation model to target speed change from $\omega = 1900$ to 2000 rpm at time 3273.5 s in one of the animal experiments (right panel). Solid line is measured flow (Q_{meas}), while dashed line shows estimated pulsatile flow (Q_{est}).

Dynamical Modelling Results (Cont'd)

- Figure 6 illustrates the linear regression analysis between $Q_{est}(k)$ and $Q_{meas}(k)$ obtained from animals $(N = 6)$. Correlation coefficient (R^2), slope of the linear regression line (l) and offset value (α) are shown in table 1 for each animal. Average correlation between estimated and measured flows was $(R^2 = 0.849)$, with small mean absolute error $(e = 0.584\ L/min)$.

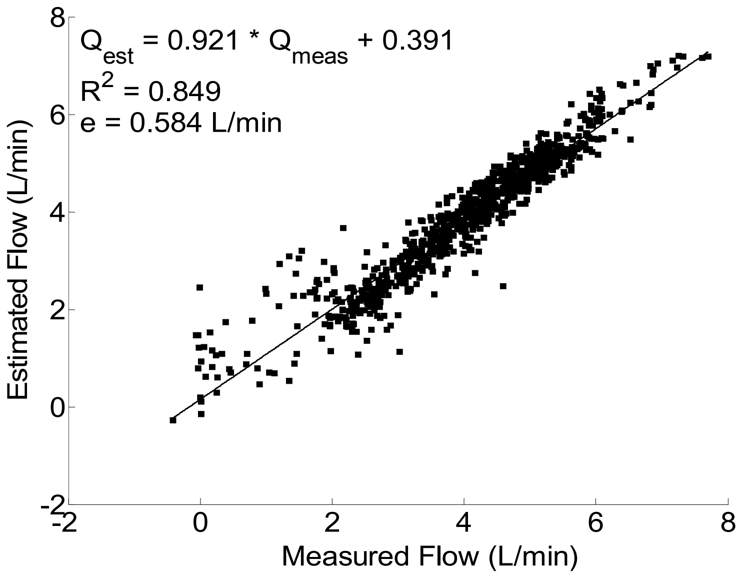


Figure (6): Linear regression plot between estimated (Q_{est}) versus measured (Q_{meas}) flow obtained from animal data $(N = 6)$.

Dynamical Modelling Results (Cont'd)

Experiment	Parameter				
	R^2	l	α	Flow range (min-max) (L/min)	e (L/min)
Pig 1	0.897	0.951	0.269	-0.5 – 7	0.324
Pig 2	0.855	0.872	0.515	-0.7 – 7	0.343
Pig 3	0.691	0.883	0.395	-3.5 – 8	1.193
Pig 4	0.935	0.953	-0.056	-0.7 – 7	0.299
Pig 5	0.794	0.891	0.532	-2.5 – 8	1.043
Pig 6	0.921	0.974	0.696	-0.5 – 7	0.302
All Animals	0.849 $\pm$0.038	0.921 $\pm$0.018	0.391 $\pm$0.107	-3.5 – 8	0.584 $\pm$0.170

Table (1): Linear regression analysis results between estimated and measured flow from six acute pigs experiments. Results for all animals show mean$\pm$sem where appropriate.

- The smaller correlation in two data sets (pig 3 and pig 5) and relatively high error between estimated and measured flow was due to the excessively noisy nature of the power signals in these data sets where they do not continue to follow the fluctuations of the low and high flow ranges rather they intermittently reach minimum and maximum plateau values.

Dynamical Modelling Results (Cont'd)

❖Head:

- The head estimation model with $n = 2, m = 4,$ and $delay\ value = 1$ that produced the highest correlation coefficient (R^2) and least mean absolute error (e) between $H_{est}(k)$ and $H_{meas}(k)$ is defined as follows [2]:

$$C(\nabla)H_{est}(k) = D_1(\nabla)Q_{est}(k) + D_2(\nabla)\omega(k) + \lambda \quad (16)$$

where

$$C(\nabla) = 1 - 1.735\nabla^{-1} + 0.758\nabla^{-2} \quad (17)$$

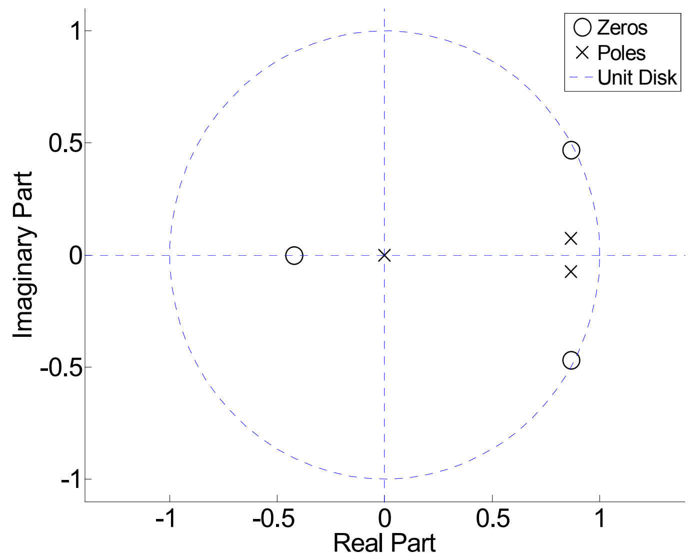
$$D_1(\nabla) = -1.157\nabla^{-1} + 1.519\nabla^{-2} - 0.278\nabla^{-3} - 0.475\nabla^{-4} \quad (18)$$

$$D_2(\nabla) = 0.0683\nabla^{-1} - 0.0742\nabla^{-2} - 0.0298\nabla^{-3} + 0.0379\nabla^{-4} \quad (19)$$

- Here $H_{est}(k)$ is the estimated instantaneous head, $Q_{est}(k)$ is the estimated pulsatile flow, $\omega(k)$ is the instantaneous rotational speed, $\lambda = -0.476$ is a constant value, and is the discrete time backward shift operator.
- Figure 7 (a) and (b) shows the poles and transmission zeros plot of the head estimation system.

Dynamical Modelling Results (Cont'd)

(a)



(b)

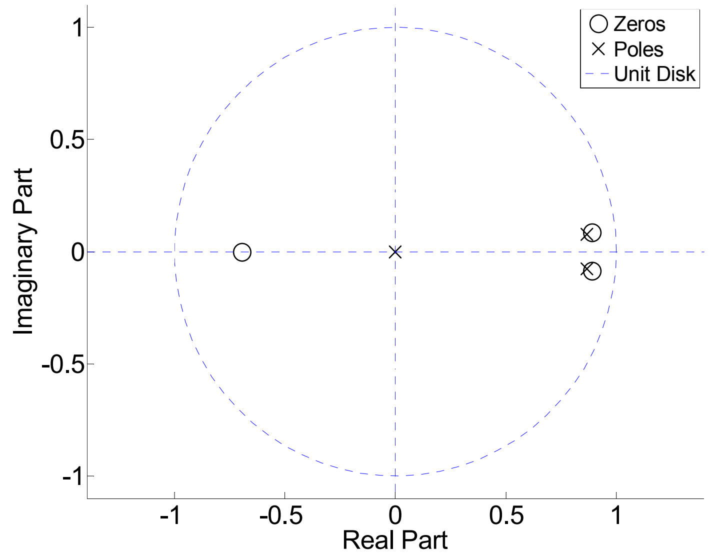


Figure (7): Poles and transmission zeros plot of the head estimation (H_{est}) model from **(a)** $Q_{est}(k)$ and **(b)** $\omega(k)$.

- It is noticeable that the proposed model is stable as all poles are inside the unit circle. As the value of delay between input (s) and output signals increased, the pulsatile flow and head estimation models became unstable. It should be pointed out that unlike pump flow model (15); the proposed equation (16) for head estimation has the constant term which makes this model nonlinear. There is a 5% difference in the absolute value between poles and transmission zeros shown in figure 7(b). Model reduction was not possible here because of the constant value in the system model in equation (16).

Dynamical Modelling Results (Cont'd)

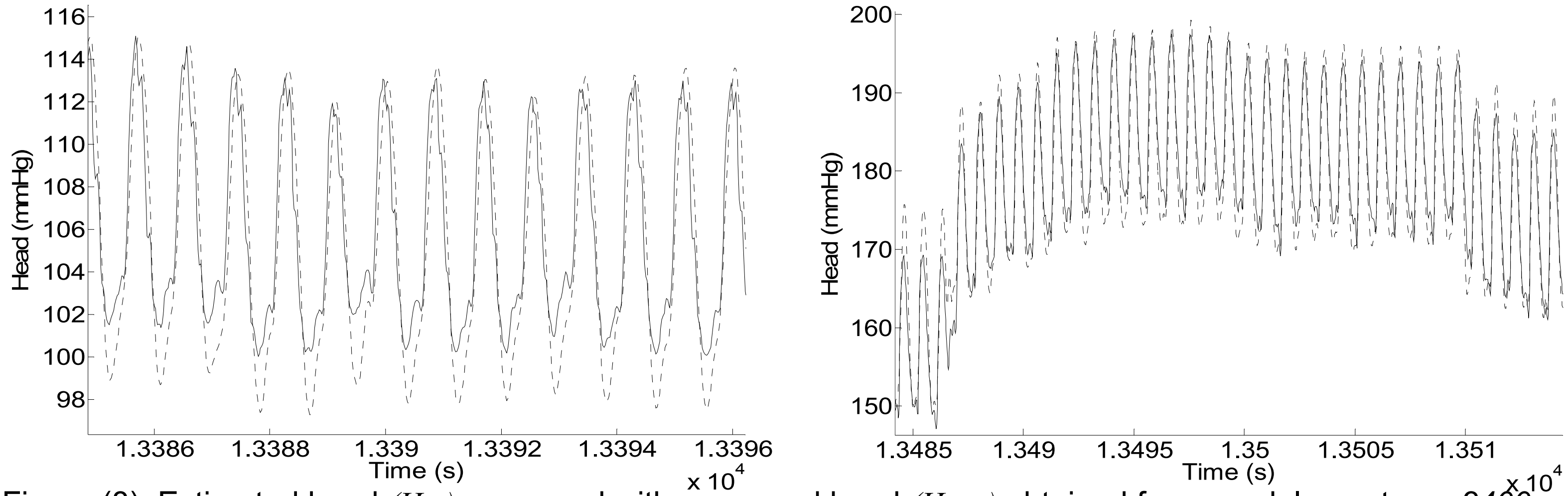


Figure (8): Estimated head (H_{est}) compared with measured head (H_{meas}) obtained from mock loop at ω = 2400 *rpm* and an *HCT of 37%* equivalent to a viscosity value of *2.95 mPas* (left panel). Transient response of the head estimation model to target speed changes from $\omega = 2900$ *rpm* to *3000 rpm* at time *13486 s* and from $\omega = 3000$ *rpm* back to *2900 rpm* at time *13509 s* with *HCT = 27%* (equivalent to viscosity value of *2.35 mPas*) (right panel). Solid line is measured head (H_{meas}), while dashed line shows estimated head (H_{est}).

- Linear regression analysis between H_{est} and H_{meas} using mock loop data is shown in figure 9(a-c). Overall, a mean absolute error $(e = 7.682\ mmHg)$ and a highly significant correlation $(R^2 = 0.933)$ were obtained.

Dynamical Modelling Results (Cont'd)

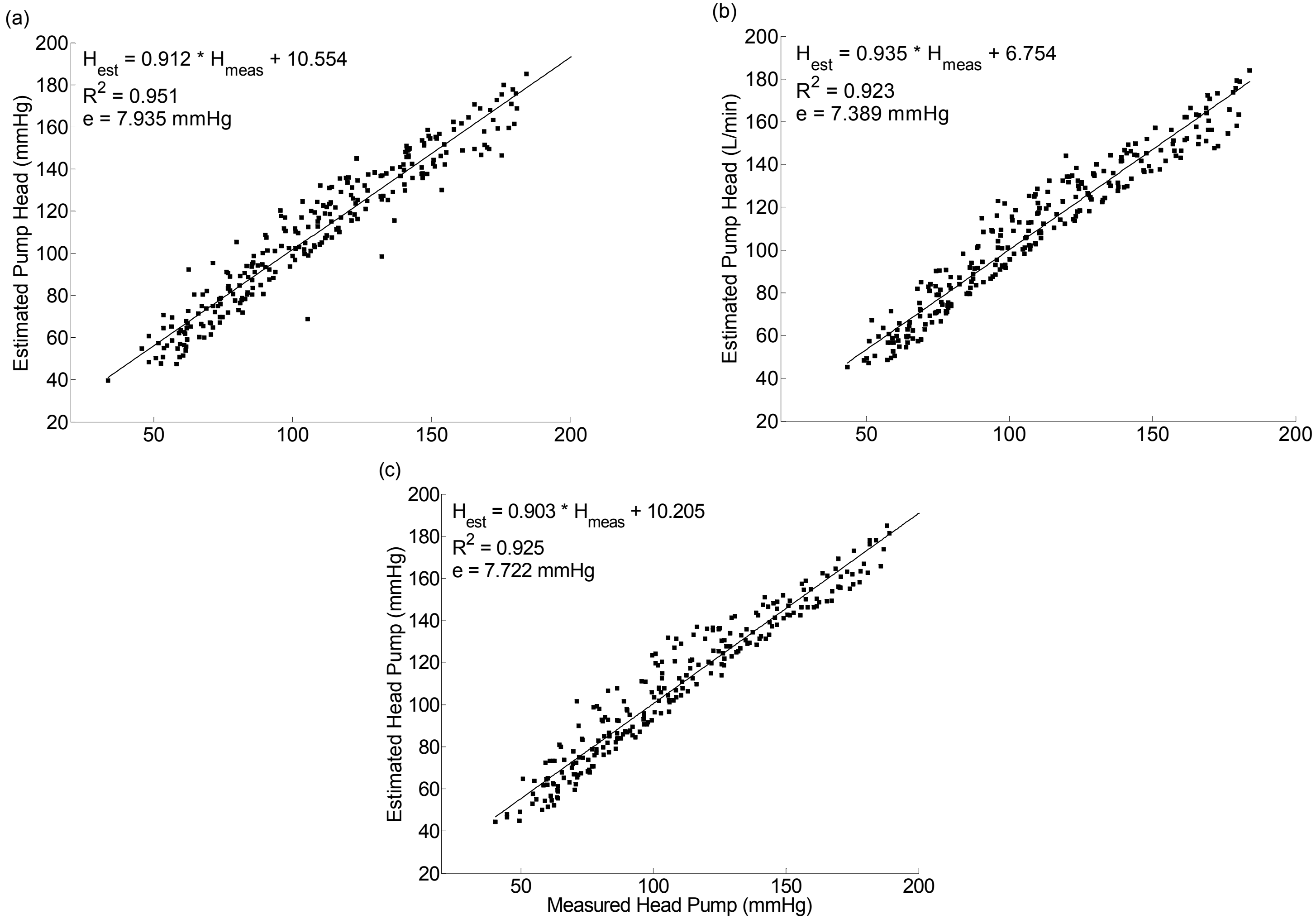


Figure (9): (a) *27 % of HCT*, (b) *37 %* and (c) *47.5 %*. The resulting slopes of linear regression lines are *l = 0.912, 0.935, 0.903* respectively.

Dynamical Modelling Results (Cont'd)

-The model can provide an accurate estimate for the negative flow up to *-0.7 L/min* as shown in figure 10. Figure 10 also shows the pulsatile flow estimation model response in the negative flow region, where the power signal becomes increasingly noisy.

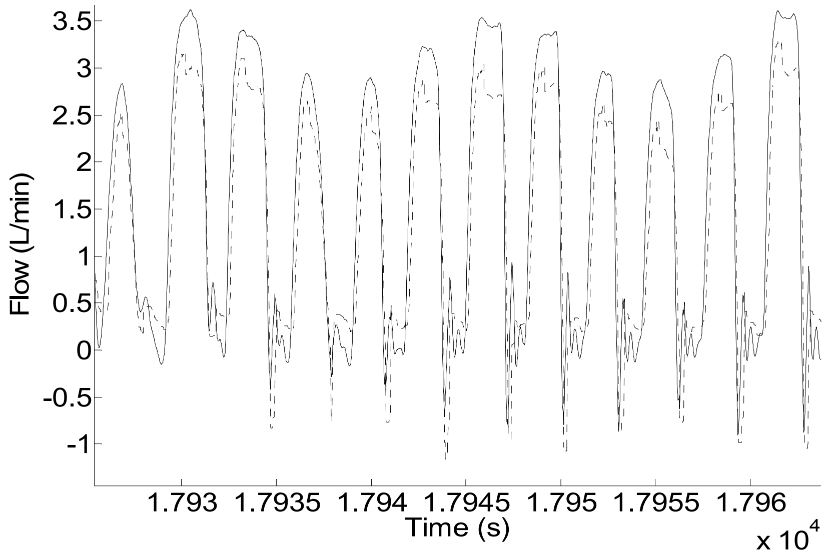


Figure (10): Estimated and measured flow in negative region up to -1 L/min with $\omega = 1800\ rpm$ in one of the animal experiments. Solid line is the measured flow, while the dashed line shows estimated pulsatile flow. Note that the model is able to accurately estimate the pulsatile flow in negative regions up *to -0.7 L/min*.

Dynamical Modelling Results (Cont'd)

-The responses of the proposed model to abnormal flow conditions associated with different pumping states such as ventricular collapse (VC) and aortic valve not opening (ANO) are shown in figures 11. As shown, the model accurately predicts the flow during these two states.

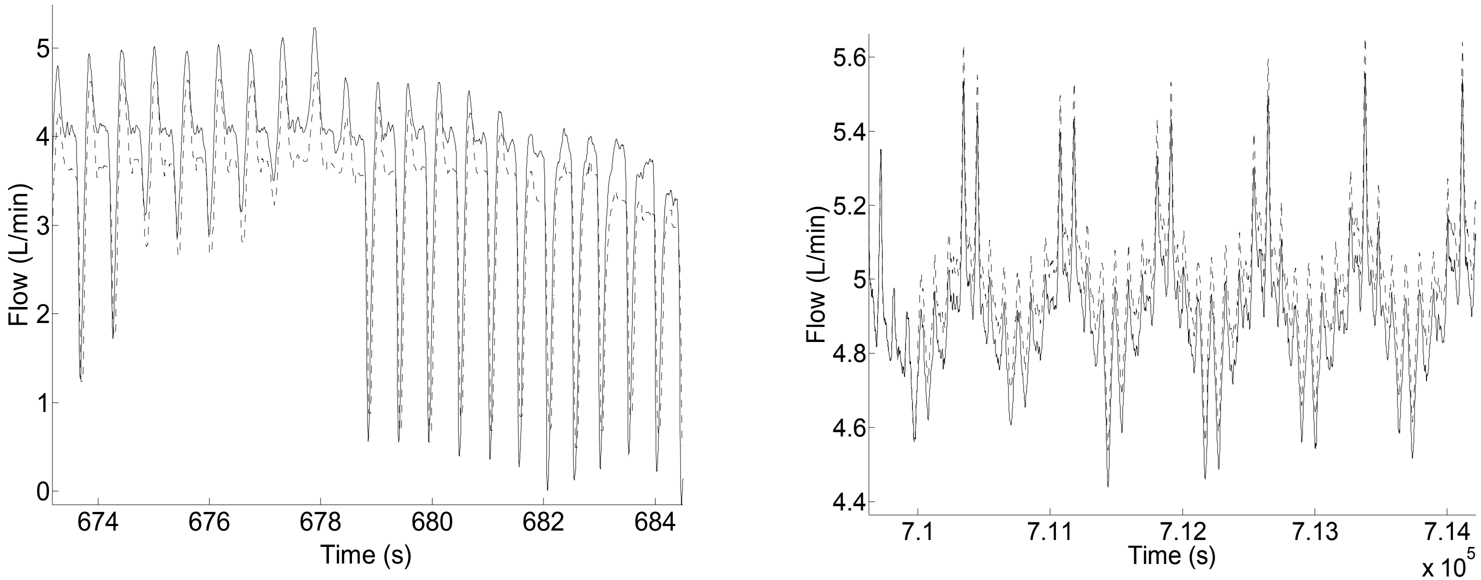


Figure (11): Estimated and measured flow. (left): VC, and (right): AVNO.

- The resulting proposed model for the head estimation of the pump was not validated using the animal experiments because accurate measurement of pump outflow pressure was not obtained. Therefore, the head estimation model was validated using the mock circulatory loop data only.

Non-invasive deadbeat controller design for an IRBP

A) Model of the ventricular assisted circulatory system

The Cardiovascular system (CVS) model:

- A software model incorporating a lumped-parameter model of cardiovascular system (CVS), developed in [5], shown in fig. 12, was used to evaluate the proposed controller algorithm.

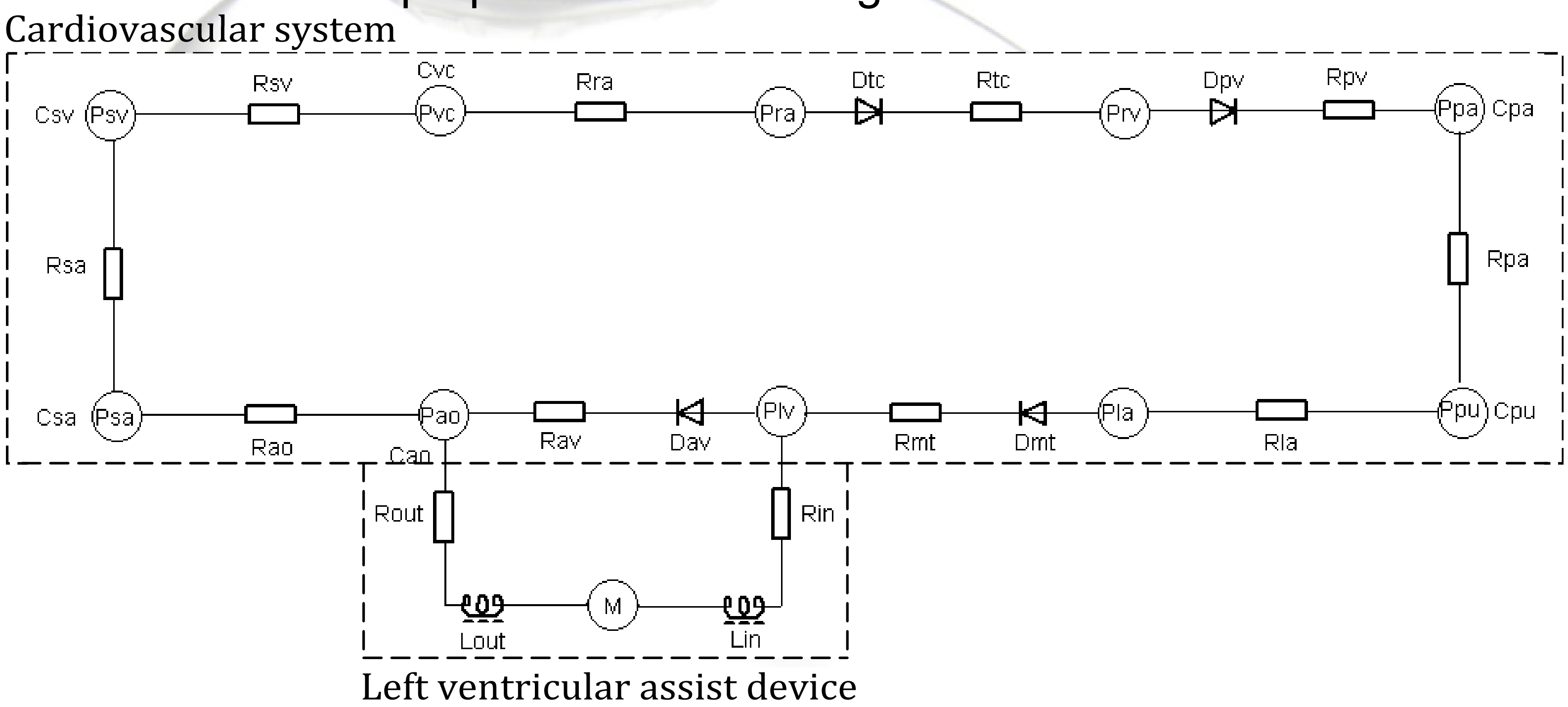


Figure (12): Schematic diagram of the ventricular assisted circulatory system model.

Materials and Methods (Cont'd)

- The model of the CVS comprised ten compartments including the left and right sides of the heart, as well as the pulmonary and systemic circulations.

- Each compartment in the CVS model was formulated based on a well established experimental observations.

- The capacitive elements (C_i) represent the compliance of the various compartments, while the resistive elements (R_i) represent the resistances between two compartments.

- Model parameters associated with left ventricular failure (LVF), including contractility of LV, systemic vascular resistance, total blood volume and heart rate were then modified to allow simulations of moderate and severe LVF conditions.

Materials and Methods (Cont'd)

- The CVS model has been carefully validated using published data from the literature, as well as animal experiments (pigs Experiments).

- A detailed description of the model with parameters values can be obtained from [5].

The LVAD model

- The LVAD model included the description of the rotary blood pump, as well as the inlet and outlet cannulae.
- The proposed stable dynamical models for pulsatile flow and head estimation of the IRBP were used to represent the pump model.
- The inlet and outlet cannulae formed an important link between the CVS model and the LVAD model.
- They were each modeled in terms of flow-dependent resistance (R_{in} and R_{out}), as well as the inlet and outlet inertances (L_{in} and L_{out}).

Materials and Methods (Cont'd)

B) Implementation of deadbeat control algorithm

- A deadbeat controller algorithm was implemented to control the pump flow.
- The ability to respond quickly to sudden perturbations in the cardiovascular system (CVS) and adjust pump flow accordingly is highly desirable due to the fact that sudden changes in afterload or preload may cause undesirable effect on CVS, such as ventricular collapse, if the pump controller does not react fast enough.
- Recall that the estimated instantaneous pulsatile flow is defined as follows [2,4]:

$$\begin{aligned} Q_p(kh) &= 0.2710\, f([k-1]h) - 0.2546\, f([k-2]h) \\ &\quad -1.985 Q_p([k-1]h) - 1.240 Q_p([k-2]h) \\ &\quad -0.2397\, Q_p([k-3]h) + e_1(kh). \end{aligned} \tag{20}$$

- Here, h is the sampling interval equal to *0.02s* and $e_1(kh)$ represents the model error.

Materials and Methods (Cont'd)

- The estimated pulsatile flow (Q_p) in (20), together with the rotational speed (ω), was used to estimate instantaneous head across the pump (H_p) as defined by [2,4]

$$
\begin{aligned}
H_p(kh) = &-0.476 - 1.157\,Q_p([k-1]h) \\
&+1.519\,Q_p([k-2]h) - 0.278\,Q_p([k-3]h) \\
&-0.475\,Q_p([k-4]h) + 0.0683\,\omega([k-1]h) \\
&-0.0742\,\omega([k-2]h) - 0.0298\,\omega([k-3]h) \\
&+0.0379\,\omega([k-4]h) - 1.735\,H_p([k-1]h) \\
&+0.758\,H_p([k-2]h) + e_2(kh). \qquad (21)
\end{aligned}
$$

- The combination between the CVS and LVAD models was performed using the inlet and outlet cannulae model which was obtained in terms of flow-dependent resistance and the inlet and outlet inertances as follows [5]:

Materials and Methods (Cont'd)

$$H_p(t) = H_{cvs}(t) + (R_{in} + R_{out})Q_p^{\,2}(t) + (L_{in} + L_{out})\dot{Q}_p(t) + e_3(t). \quad (22)$$

- Here, $H_p(t)$ denotes the differential pressure across the pump, $H_{cvs}(t)$ denotes pressure difference between the left ventricle and the aorta, and $e_3(t)$ is the model error.

- Parameters of the cannulae model were estimated using data from the in vivo animal experiments [5].

- To obtain CVS-LVAD model, equation (22) was discretized to obtain a model for the pump differential pressure [4]:

$$H_p(kh) = H_{cvs}(kh) + 1.67Q_p^{\,2}(kh) + \frac{0.83}{h}(Q_p(kh) - Q_p([k-1]h)) + e_3(kh). \quad (23)$$

Materials and Methods (Cont'd)

- Consider the control system shown in fig. 13, with the pulsatile pump flow (Q_p) as the control variable [4].

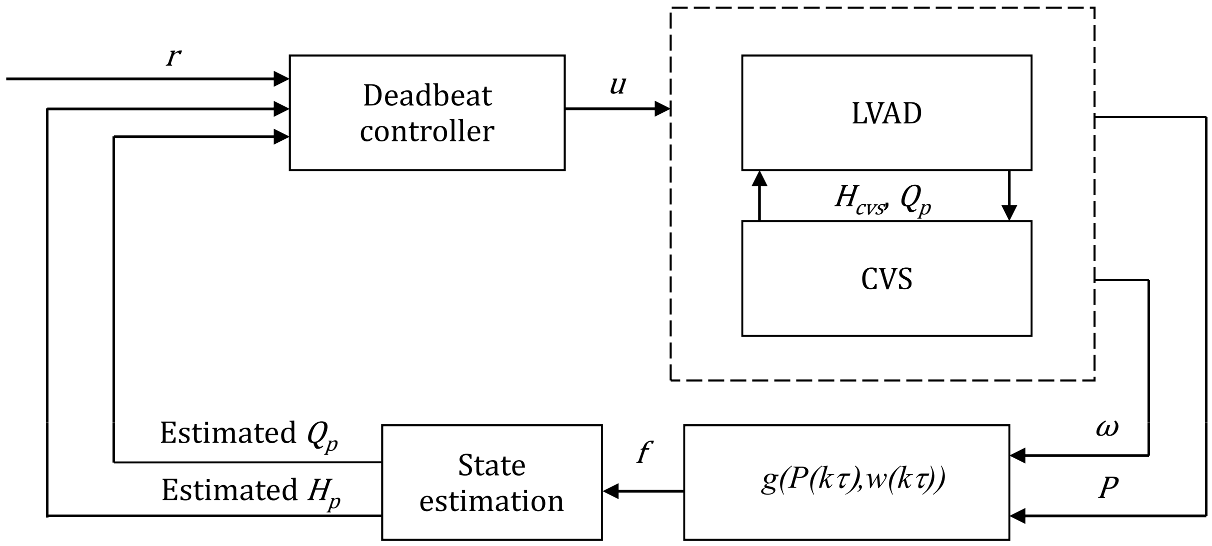


Figure (13): Block diagram of the deadbeat control systems.

- The inputs to the deadbeat controller are the reference pump flow, the estimated pump flow and the estimated head, while the output of the controller is the pulse width modulation signal *(PWM)* voltage signal, which is the actual control input to the pump, represented as u.

Materials and Methods (Cont'd)

- Pump rotational speed *(ω)* and power *(P)* were the only variables measured in the system, and estimated steady state flow *f(.)* were obtained using (1).
- *f(.)* was then used to estimate the pulsatile flow and then the pump head using (20) and (21).
- The inputs to the deadbeat controller are the reference pump flow, the estimated pump flow and the estimated head, while the output of the controller is the pulse width modulation signal *(PWM)*, which is the actual control input to the pump, represented as *u(.).*
- New ARX model with four input signals which describes the relationship between the control signals i.e. the *PWM* signal, *u(.),* the steady state pump flow, *f(.),* the pulsatile pump flow, $Q_p(.)$, and the pump head, $H_p(.)$ was obtained using the mock loop data as follows [4]:

$$f(kh) = d_1 u([k-3]h) + d_2 f([k-1]h) - d_3 f([k-2]h) + d_4 Qp([k-3]h) \quad (24)$$
$$+ d_5 H_p([k-3]h) + e_4(kh).$$

Materials and Methods (Cont'd)

- Here, d_1, d_2, d_3, d_4, and d_5 are constants with values of 0.004199, 1.956, -0.962, 0.05766, and -0.0005538 respectively and $e_4(k)$ is the model error.
- Correlation between measured and estimated *f(.)* was highly significant (R^2 = *0.945*), and the slope of the linear regression line was very close to the unity (*slope = 0.961*).

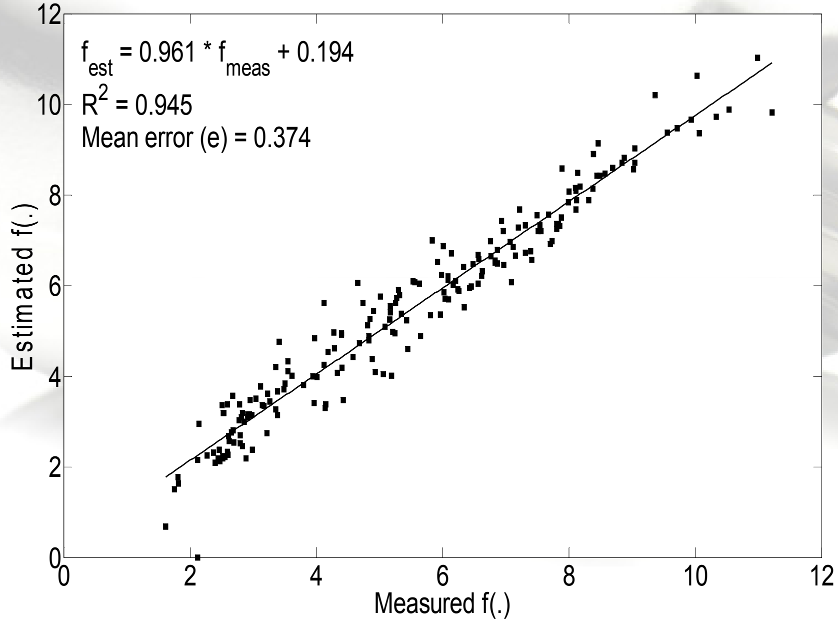


Figure (14): Estimated steady state pump flow versus measured one.

Materials and Methods (Cont'd)

- In the system, the sampled $H_{cvs}(kh)$ in equation (23) is obtained from the CVS model.
- In the present simulation, two cases were considered: (i) constant reference input, i.e. $r(t) = a$, where $a = constant > 0$, and (ii) sinusoidal reference input, i.e $r(t) = a+b\sin(\omega t)$, where a and b are constants, $a \geq b$.
- Consider the following deadbeat controller [4]:

$$\overline{f}(k\tau) = \frac{1}{0.271}\begin{pmatrix} 0.2397\,Q_p([k-2]\tau) + 1.240\,Q_p([k-1]\tau) + 1.985\,Q_p(k\tau) \\ +\,0.2546\,f([k-1]\tau) + r([k+1]\tau) \end{pmatrix}, \quad (25)$$

$$u(k\tau) = \frac{1}{d_1}\begin{pmatrix} \overline{f}([k+3]\tau) - d_2 f([k+2]\tau) + d_3 f([k+1]\tau) - d_4 Q_p(k\tau) - \\ d_5 H_p(k\tau) \end{pmatrix}. \quad (26)$$

- Now, solve equations (20) and (24) with initial conditions.
- Substitute $H_{cvs}(kh)$ in (23) to obtain $H_p(kh)$, and then substitute into (24).

Materials and Methods (Cont'd)

- Now, apply controller (25) and (26) on (24) to obtain $Q_p(kh) = r(t)$ after certain number of samples.
- In all simulations, we added a sinusoidal high frequency signals to the model error terms in equations (20), (21), (22), and (23).
- The simulation was first carried out using varying levels of constant reference pump flow input to study the hemodynamic response of the CVS under various heart conditions, i.e. (i) healthy, (ii) moderate LVF and (iii) severe LVF.
- Model parameters associated with LVF, including contractility of LV and RV, systemic vascular resistance, total blood volume, and heart rate were carefully chosen in order to ensure that realistic simulation, in terms of CO, P_{ao}, P_{lv}, was achieved.
- Next, sinusoidal signals of varying mean values, amplitude, and phase shift were applied to the reference pump flow input.
- In all simulations, frequencies of the sinusoidal signals were chosen to be equal to the heart rates.

Deadbeat Controller Simulation Results

(i) Constant reference input:

- The simulation was first carried out using constant reference pump flow input, starting with 2.5 L/min at *t* = 0 s, and linearly increased to 5 L/min from *t* = 10s to 15s, using parameters for severe LVF condition.

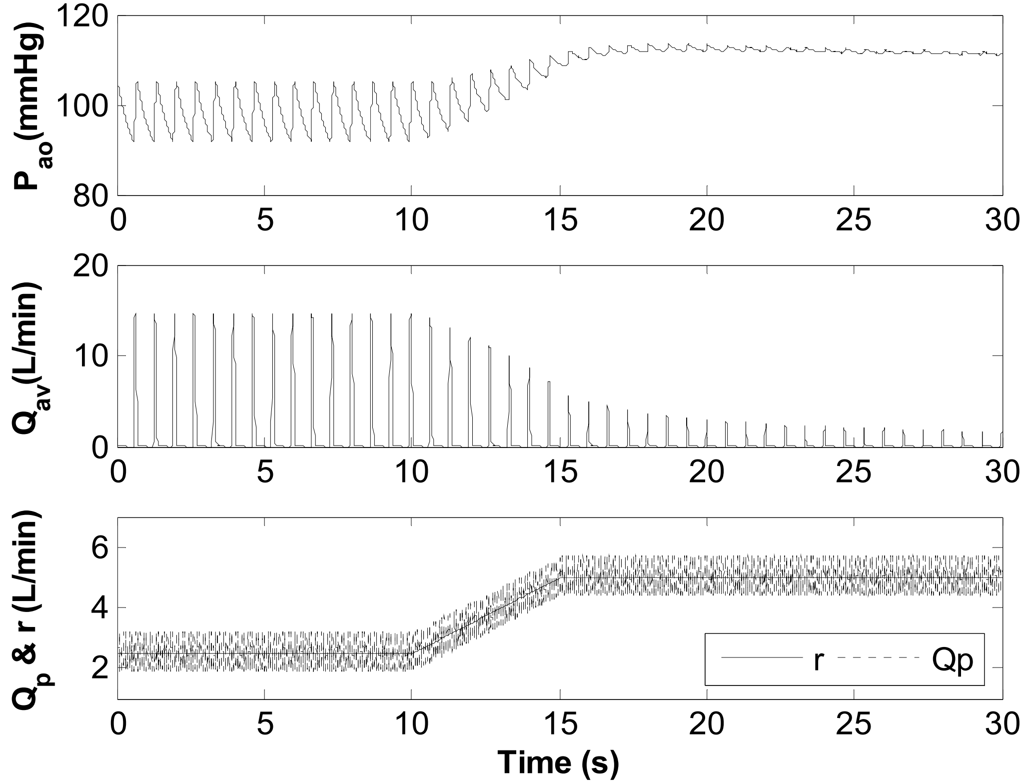


Figure (15): *Aortic pressure (P_{ao}), aortic valve flow (Q_{av}), pump flow (Q_p) and reference pump flow (r) waveforms from the model simulations using parameters for severe LVF condition. Reference pump flow input was increased linearly from 2.5 L/min at t = 10s to 5 L/min at t = 15s.*

Deadbeat Controller Simulation Results (Cont'd)

- Fig. 16 shows the trend of cardiac output (*CO*), mean aortic valve flow (mQ_{av}), mean aortic pressure (mP_{ao}) and mean left atrial pressure (mP_{la}) with increasing pump flow under three different heart conditions, i.e. (i) healthy, (ii) moderate LVF and (iii) severe LVF.

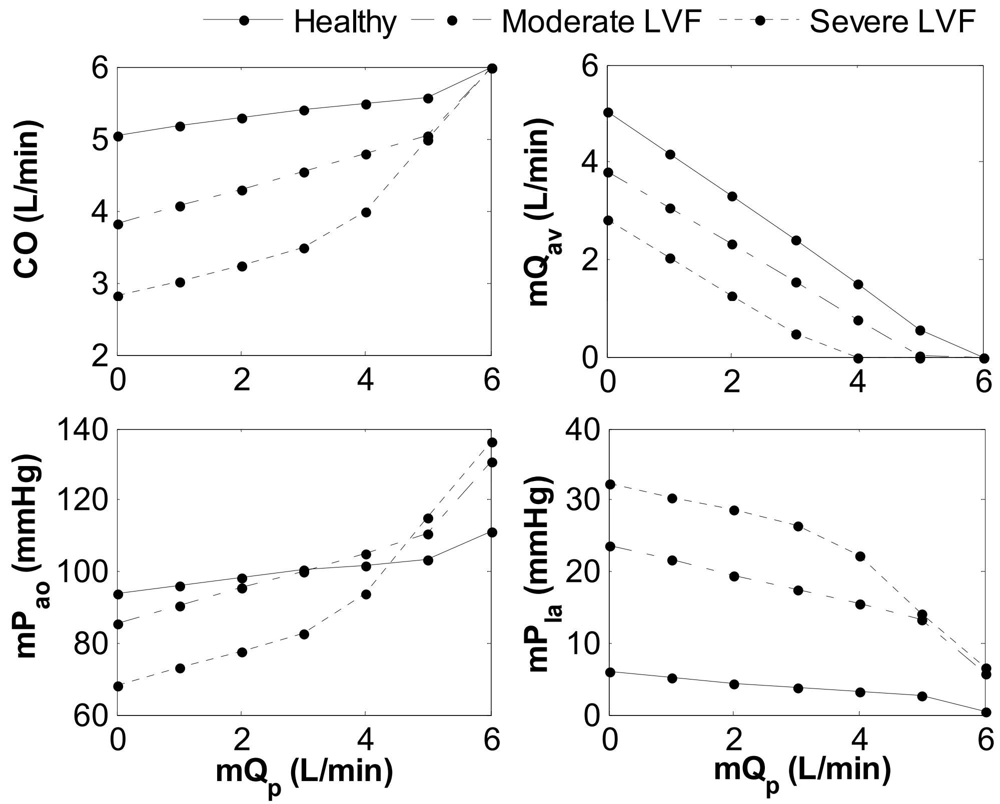


Figure (16): *Effect of mean pump flow (mQ_p) on total cardiac output (CO), mean aortic valve flow (mQ_{av}), mean aortic pressure (mP_{ao}) and mean left atrial pressure (mP_{la}) under three different heart conditions: healthy, moderate left ventricular failure (LVF) and severe LVF.*

Deadbeat Controller Simulation Results (Cont'd)

- Increasing the severity of LVF decreases cardiac output and mean arterial pressure while increasing left atrial pressure, due to blood volume pooling in the pulmonary circulation.

- In all three cases of varying heart conditions, increasing pump flow decreases left atrial pressure, leading to a decrease in stroke volume and aortic valve flow through the Frank-Starling mechanism. This is consistent with published findings.

- On the other hand, improvement in total cardiac output and mean arterial pressure with increasing pump flow is most apparent in severe LVF patients, followed by moderate LVF patients and healthy subjects, as observed experimentally.

Deadbeat Controller Simulation Results (Cont'd)

(ii) Sinusoidal reference input:

- Fig. 15 shows the waveforms of the aortic pressure (P_{ao}), aortic valve flow (Q_{av}) and pump flow (Q_p) superimposed on the reference pump flow signal (r) generated by the model using parameters for severe LVF condition. Reference input signal is increased from $r(t) = 2.5 + 1.5\sin\omega t$ to $r(t) = 5 + 1.5\sin\omega t$ at $t = 10s$. It is shown that the simulated pump flow accurately tracked the reference input signal within an error of ± 0.7 L/min.

- Next, total blood volume was decreased linearly from 5100 mL to 4600 mL from t = 20s to 30s using model parameters for moderate LVF to evaluate the performance of the controller under perturbation of model parameters. The controller successfully maintained the desired pump flow signal, $r(t) = 4 + 2\sin\omega t$ during this perturbation (Fig. 16).

Deadbeat Controller Simulation Results (Cont'd)

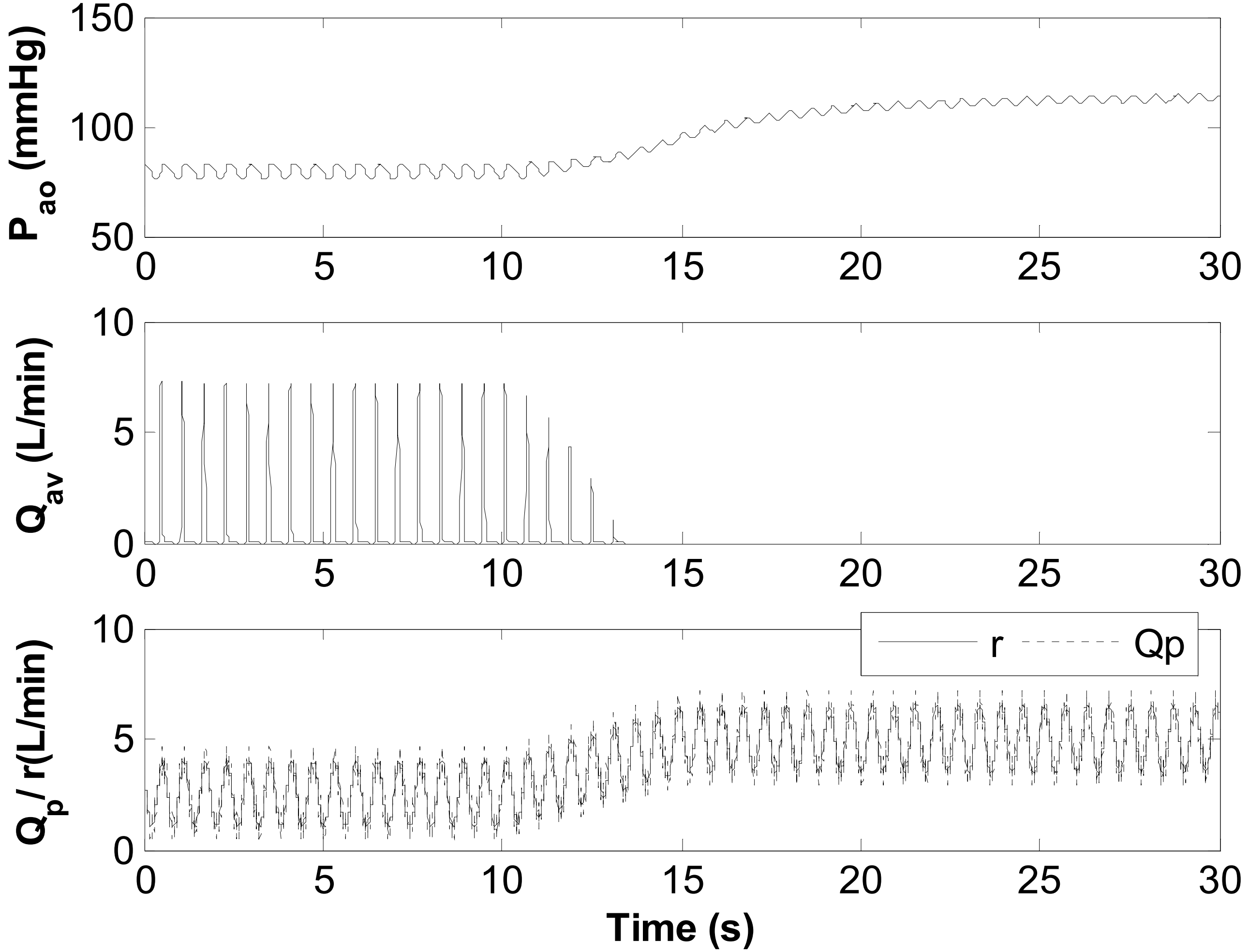


Figure (15): Aortic pressure (P_{ao}), aortic valve flow (Q_{av}), pump flow (Q_p) and reference pump flow (r) waveforms from the model simulations using parameters for severe LVF condition. Reference pump flow input was increased linearly from 2.5+1.5sinωt L/min at t = 10s to 5+1.5sinωt L/min at t = 15s.

Deadbeat Controller Simulation Results (Cont'd)

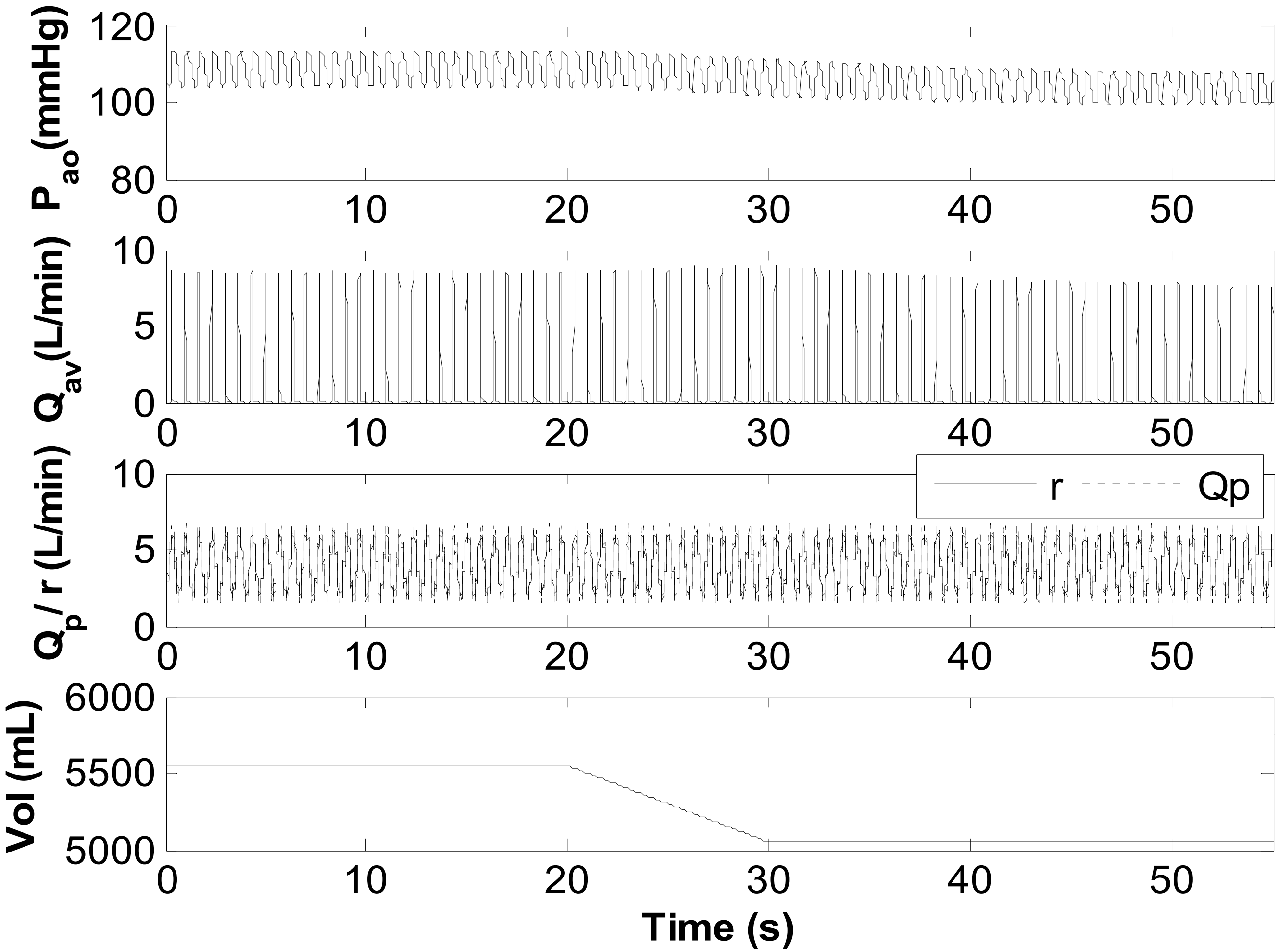


Figure (16): Aortic pressure (P_{ao}), aortic valve flow (Q_{av}), pump flow (Q_p), reference pump flow (r) and total blood volume (Vol) waveforms from the model simulations using parameters for moderate LVF condition. Total blood volume was decreased linearly from 5100 mL at t = 20s to 4600 mL at t = 30s.

Conclusions

- Two stable dynamical models were proposed for non-invasive estimation of pulsatile flow and head in an IRBP.
- Models were validated using circulatory mock loop and animals data.
- Models are helpful in studying the transient response of the pump flow and pressure.
- Performance of a deadbeat controller was examined and showed its ability to track the reference inputs with minimal error in the presence of model uncertainty by using non-invasive measurements of rotational speed and the power of the pump.